

Professional Self-Assessment – Gary Travis

I am a computer science student with experience in software development, databases, and programming across multiple project types, and this ePortfolio presents work that demonstrates my growth throughout the program. Through the enhancements in this portfolio, I focused on improving the quality, security, and maintainability of real projects rather than creating examples that only met minimum assignment requirements. The artifacts included here show how I apply computing concepts in a practical way and how my technical abilities have developed over time.

Over the course of the program, I developed stronger software design and engineering skills, especially in organizing code into clearer structures, reducing duplication, and improving maintainability. I also strengthened my understanding of algorithms and data structures by learning how to choose structures and logic that better support program behavior, efficiency, and readability. In addition, I gained practical experience working with databases and data-driven applications, including creating database connections, designing queries, handling data securely, and supporting interactive application behavior.

My problem-solving and debugging skills also improved significantly during the program. I became more comfortable reading error messages, tracing issues back to their actual cause, and revising code carefully instead of relying on trial and error. I also improved my written and technical communication skills by documenting my design decisions, writing clearer comments, and explaining enhancements through the narratives included in this portfolio.

The enhanced CS330 artifact demonstrates my growth in algorithms and data structures. For this project, I improved a 3D scene application by changing it from a long sequence of hard-coded draw calls into a more organized, data-driven scene graph structure. This enhancement allowed

me to demonstrate the use of structured data, recursive logic, and cleaner rendering organization. It also showed my ability to evaluate design trade-offs and improve a project in a way that made it easier to maintain and extend.

The enhanced CS340 artifact demonstrates my growth in software design and engineering as well as databases. In that work, I improved a Python CRUD module and dashboard application connected to a MongoDB database. I refactored repeated logic into helper functions, improved validation and error handling, and made the dashboard structure easier to read and maintain. I also strengthened the database portion of the artifact by improving authentication, removing hard-coded credentials, adding protections against unsafe query behavior, and using database features such as indexing and aggregation to improve efficiency and reliability.

One of the most important things I learned during the computer science program was that successful software is not only about whether it runs correctly, but also about whether it is secure, maintainable, and understandable to others. Revisiting earlier work for this ePortfolio helped me recognize weaknesses that were not obvious when I originally completed the assignments. That process gave me a better understanding of professional expectations and helped me grow from simply completing projects to thoughtfully improving them.

This ePortfolio also reflects my growth in connecting technical work to broader course outcomes. I now have a better understanding of how software engineering principles, algorithmic thinking, database design, and communication all work together in real computing practice. The enhancements in this portfolio are not just revisions; they are examples of how I can review existing work, identify weaknesses, and make targeted improvements that add value.

As I move forward professionally, I plan to pursue opportunities in software development where I can continue building my skills in programming, backend development, and data-driven systems. I want to continue learning more about testing, secure development practices, and modern development environments so I can contribute effectively to production-quality software. This ePortfolio represents both what I have learned in the program and the direction I want to continue pursuing in my career.